

- **Problem Statement ID** – [Enter PS ID]
- **Problem Statement Title** – Automated Road Obstruction & Hazard Mapping from Street/Dashcam Video
- **Team Name** – Team UniVerse
- **Team Leader Name** – [Enter team leader name]
- **Members Names** – [Enter member names]

IDEA TITLE: HazardMap

Proposed solution: any phone or dashcam becomes a road-hazard sensor.



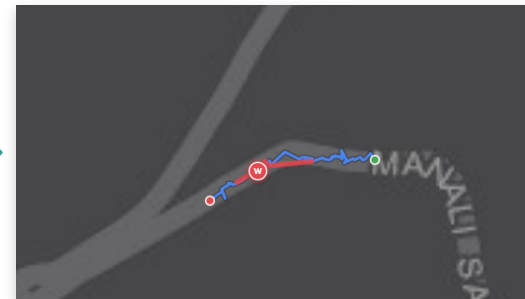
1 Camera sees the road

Helmet/dash camera on NH3, Himachal



2 AI spots the hazard

Water flowing across road: flooding



3 Pinned on the map

High-severity line, exact GPS location



Sees what satellites miss

Fallen trees, wires, waterlogging under canopies & overpasses



Exact hazard location

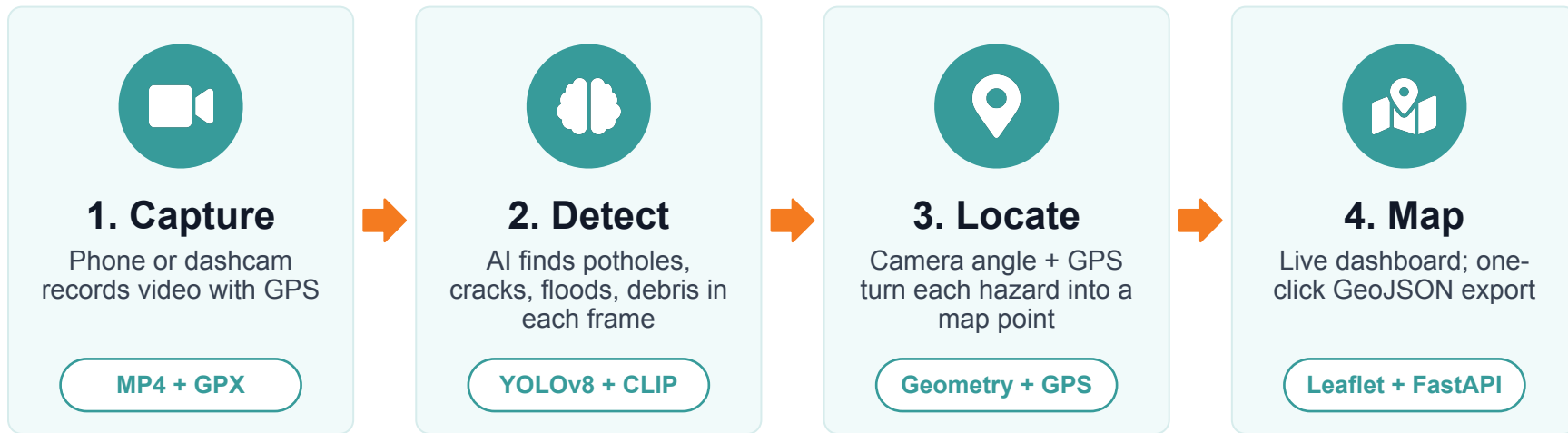
Camera geometry pins the hazard itself, not just the car



Live & standard

Real-time map for responders + GeoJSON for any GIS

How it works — in four steps



Built with

Python

YOLOv8

YOLO-World

CLIP

OpenCV

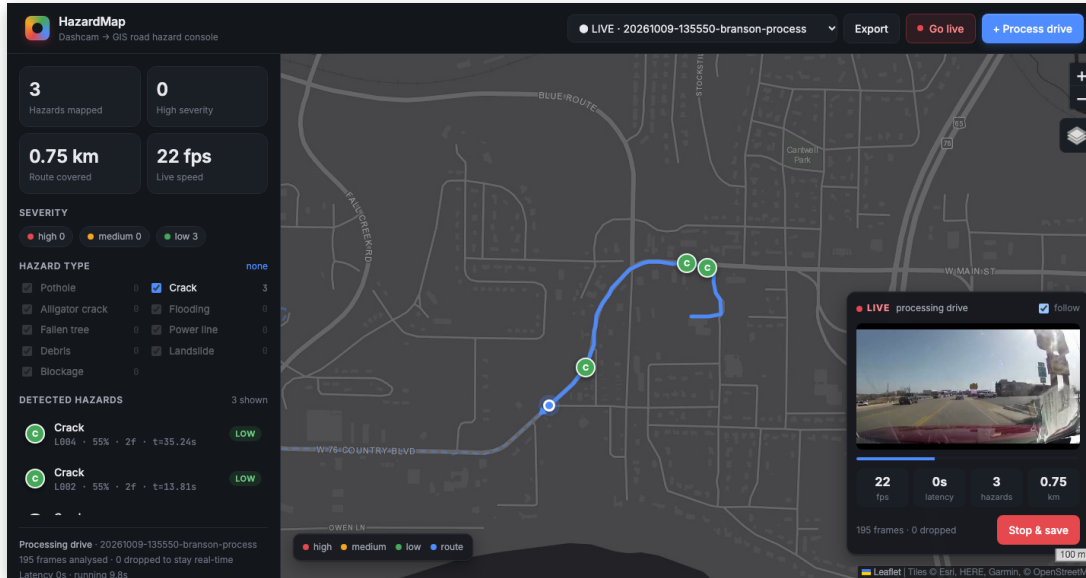
FastAPI

Leaflet

GeoJSON



Runs on a laptop GPU (Apple M5) — no cloud needed. Three ways in: upload a drive, replay it live, or stream a phone camera.



Live view while a drive is processed: the vehicle moves along the route and hazards appear on the map as they are found.

22 fps AI speed on a laptop GPU

4× faster than real time

3 inputs: upload · live replay · phone



Evidence photo saved with every hazard

Feasibility analysis



Already working end-to-end
Prototype built & tested, incl. live mode and a phone-as-dashcam



No special hardware
Runs on a laptop GPU; any phone/dashcam is the sensor



Open data & models
Pretrained on RDD2022 (47k+ images); open-source stack, zero licence cost

Challenges & risks → our strategies

Challenge / risk	Strategy
AI misses some hazards (trees, debris)	Fine-tune on labelled disaster photos; a CLIP double-check already stops false flood alarms on dry roads
Intermittent / noisy GPS	Time interpolation + heading smoothing; gap flag per hazard; road map-matching next
Unknown camera mounting	Configurable height / pitch / FOV; auto-calibration from horizon & lane lines
Same hazard seen in many frames	ByteTrack identities + DBSCAN clustering → one feature per real hazard



Municipal corporations

Continuous pothole & road-damage inventory from buses, garbage trucks, taxis — prioritise repairs by severity



Disaster response teams

Within hours of a storm or landslide: which roads are blocked, flooded or hit by fallen lines



Citizens & commuters

Safer routing, fewer accidents and vehicle damage; transparent public hazard map

Key benefits

- **Low cost** — reuses phones & dashcams already on the road
- **Fast** — hazard map in minutes, not days of surveys
- **Interoperable** — standard GeoJSON for any GIS or city dashboard
- **Scalable** — every vehicle adds coverage; repeat drives confirm



Tested on an Indian highway (NH3, Rohtang): flooding detected, high severity

1

RDD2022 road damage dataset

[Arya et al., arXiv:2209.08538 · github.com/sekilab/RoadDamageDetector](#)

2

comma2k19 driving dataset (video + GNSS)

[github.com/commaai/comma2k19](#)

3

YOLOv8 / YOLO-World (Ultralytics)

[docs.ultralytics.com · Cheng et al., YOLO-World, CVPR 2024](#)

4

RDD-trained YOLOv8s weights

[github.com/oracl4/RoadDamageDetection](#)

5

CLIP: zero-shot image–text matching

[Radford et al., ICML 2021 · github.com/openai/CLIP](#)

6

ByteTrack multi-object tracking

[Zhang et al., ECCV 2022](#)

7

DBSCAN density clustering

Ester et al., KDD 1996

8

GeoJSON format · Leaflet maps

[IETF RFC 7946 · leafletjs.com](#)

9

Our prototype source code

[github.com/aryan270506/Team-UniVerse-Geo-AI](#)

10

Indian road footage (NH3 demo)

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